

Ref. No.: CECONE/NPI/003

B.Sc. (Hons.) Semester- I Examination 2024-25**Statistics****STAMN-12: Practical Based on STAMN11**

Time: Three Hours

Max. Marks: 70

*NOTE: Attempt any **THREE** questions. All questions carry equal marks.*

1) Given below are the marks obtained in Mathematics by a batch of 84 students of a school:

43	34	43	32	87	35	71	65	12	52	19	48	17	24
52	65	40	54	62	45	02	13	18	49	57	21	64	71
45	81	52	40	35	78	43	45	44	55	79	37	19	14
37	71	51	35	27	74	22	08	22	15	20	23	35	27
17	21	52	48	09	22	12	12	85	40	39	30	42	08
66	74	19	04	30	06	19	31	25	38	32	51	68	43

- Summarize the data in a grouped frequency distribution using the formula given by H. A. Sturges.
- Find the proportion of students scoring (i) less than 30 marks and (ii) more than 75 marks.
- Construct a frequency histogram for the data and comment on the shape of the distribution.
- Draw the ogives and hence find the mark below which 50% of the students scored.

2) The following data represent the birth weights at an inner-city hospital in a large eastern city:

2.4, 3.3, 4.1, 5.0, 5.1, 5.2, 5.6, 5.8, 5.9, 5.9, 6.0, 6.1,
 6.2, 6.3, 6.3, 6.4, 6.4, 6.5, 6.7, 6.8, 7.2, 7.4, 7.5, 7.5,
 7.6, 7.6, 7.7, 7.8, 7.8, 7.9, 7.9, 8.3, 8.5, 8.8, 9.2, 9.7,
 9.8, 9.9, 10.0, 10.3, 10.5

- Find out first four raw and central moments for grouped data and verify the relationships among them.
- Calculate the coefficients of skewness and kurtosis and comment on the shape of the distribution.
- Draw a boxplot for the data.

3)

- Following are the marks given by two judges to the participants in a singing competition:

Judge A:	68	64	15	50	64	80	75	40	55	64
Judge B:	62	58	68	45	81	60	68	48	50	70

Obtain the rank correlation coefficient between the marks given by both the judges.

- (b) Following table represents the number of deaths from tuberculosis in England and Wales in 1956:

	Males	Females	Total
Tuberculosis of respiratory system	3,534	1,319	4,853
Other forms of tuberculosis	270	252	522
Total	3,804	1,571	5,375

Compute a measure of association for the above data.

- 4) The following bivariate frequency table represents data on the length (in cm.) of green jute plant and the weight (in gm.) of dry jute fibre of an agricultural farm in Bengal.

	Length of green plant (cm.)							Total
Weight of dry jute fibre (gm.)	111.5	127.5	143.5	159.5	175.5	191.5	207.5	
1.175	12	25	15	1				53
2.775	1	4	33	59	29	3		129
4.375	1		4	28	35	14	2	84
5.975				2	20	18	1	41
7.575				1	1	14	5	21
9.175					4	8	2	14
10.775						3	2	5
12.375							3	3
Total	14	29	52	91	89	60	15	350

- (a) Compute the coefficient of correlation between the two variables. Also show the type of association with the help of scatter diagram.
- (b) Fit the two lines of regression and calculate the angle between them.